Maximal Dissociated Subsets of Finite Abelian Groups

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Abstract

A subset A of an abelian group G is dissociated if $0 \in G$ cannot be written as the signed sum of any of the elements of A. Furthermore, the dissociativity dimension of A, denoted $\dim A$, is the maximum size of a dissociated subset of A. We are interested in evaluating the function $\dim(G, m)$, which yields the minimum value of $\dim A$ when $|A| = m$. Here, we provide some basic values and bounds for this function, and evaluate it for certain elementary abelian p-groups. In particular, we obtain exact values when the lower bound is sharp for $p = 2$ and $p = 3$. We also disprove a conjecture made by B. Bajnok in 2018 regarding the exact value for cyclic groups.

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1 Introduction

In this paper, when not specified, we will refer to G as an arbitrary finite abelian group of order n and exponent κ, and take m to be a positive integer. We begin with some definitions.

Definition 1.1. A subset $A = \{a_1, \ldots, a_m\} \subseteq G$ is dissociated if every possible equality of the form

$$\sum_{i=1}^{m} \lambda_i a_i = 0,$$

where $\lambda_i \in \{-1, 0, 1\}$, implies that $\lambda_i = 0$ for all i. We usually denote the set of all possible sums as defined above by ΣA, and write $0 \notin \Sigma A$ if A is dissociated.

Dissociated subsets of a group have been investigated, however there is an interest in finding dissociated subsets of any subset of a group. For this, we define

Definition 1.2. Let $A \subseteq G$. The quantity

$$\dim A := \max\{|D| \mid D \subseteq A,\ D \text{ is dissociated}\}$$

is known as the dissociativity dimension of A in G.

Throughout this paper, we refer to $\dim A$ as simply the dimension of A. Finally, we will define the main function that we are interested in investigating.

Definition 1.3. Given G and $m \leq n$, define

$$\dim(G, m) := \min\{\dim A \mid A \subseteq G,\ |A| = m\};$$

that is, the minimum dissociativity dimension of any subset of G of size m.

2 Previous Results

We currently have the following:

Theorem 2.1 (Lev and Luster, 2011; [1]). For all G and $A \subseteq G$, we have

$$r_A \leq \dim A \leq \lfloor r_A \cdot \log_2 \kappa \rfloor$$

where r_A is the rank of the subgroup $\langle A \rangle$ generated by A.

Proposition 2.2 (Bajnok, 2018; [2]). If G has order n and invariant factorization

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r},$$

then

$$\lfloor \log_2 n_1 \rfloor + \ldots + \lfloor \log_2 n_r \rfloor \leq \dim(G, n) \leq \lfloor \log_2 n \rfloor.$$

In particular, for all positive integers n, we have

$$\dim(\mathbb{Z}_n, n) = \lfloor \log_2 n \rfloor.$$

Proposition 2.3 (Bajnok, 2018; [2]). For all G, we have $\dim(G, 1) = 0$, $\dim(G, 2) = 1$, and

$$\dim(G, 3) = \begin{cases} 1 & \text{if } \kappa \geq 3; \\ 2 & \text{if } \kappa = 2. \end{cases}$$

3 Main Results

Elementary Results

We first evaluate the cases of $m = 4$ and $m = 5$ for all G to obtain an extension of Proposition 2.3.

Proposition 3.1. For all G, we have $\dim(G, 4) = 2$ and

$$\dim(G, 5) = \begin{cases} 2 & \text{if } \kappa \geq 3; \\ 3 & \text{if } \kappa = 2. \end{cases}$$

Proof. When $\kappa = 2$, both claims follow from Theorem 3.8. Therefore, assume $\kappa \geq 3$. For $m = 4$, let

$$A = \{0, g_1, -g_1, g_2\}$$

such that $g_1, -g_1$, and g_2 are all nonzero and distinct. For $m = 5$, a non-boolean abelian group G of order at least 5 contains at least two elements $g_1, g_2 \in G$ such that $g_1, g_2 \neq 0, -g_1, -g_2$, so we can let

$$A_1 = A \cup \{-g_2\}$$

such that $A \cap \{-g_2\} = \emptyset$. Clearly both A and A_1 do not have a dissociated subset of size 3, so $\dim A, A_1 \geq 2$. In both A and A_1, one can easily see that $0 \notin \Sigma\{g_1, g_2\}$, so $\dim A = \dim A_1 = 2$. Now, since only subsets of a set of the form $\{0, g_1, -g_1\}$ have dimension less than 2, we have $\dim(G, 4) = \dim(G, 5) = 2$ as claimed. ∎

We next introduce several bounds. This first proposition states that the dimension function is monotonically increasing over m, which is an important fact that will be used later.

Proposition 3.2. If m_1 and m_2 are positive integers such that $m_1 \geq m_2$, then

$$\dim(G, m_1) \geq \dim(G, m_2).$$

Proof. For simplicity, let $\delta_1 = \dim(G, m_1)$ and $\delta_2 = \dim(G, m_2)$. We must prove that every subset of G of size m_1 has dimension at least δ_2. Let $A \subseteq G$ be arbitrary and of size m_1. Since $|A| \geq m_2$, A contains at least one subset B of size m_2. By definition of δ_2, every subset of G of size m_2 has dimension at least δ_2, so $\dim B \geq \delta_2$. Thus B contains a dissociated subset D of size δ_2, and $B \subseteq A$, so we have $D \subseteq A$. By definition of dimension, $\dim A \geq |D| = \delta_2$ as claimed. ∎

Proposition 3.3. For all G of order n, and all $m \leq n$, we have

$$\dim(G, m) \leq \lfloor \log_2 n \rfloor.$$

Proof. By Proposition 2.2, $\dim(G, n) \leq \lfloor \log_2 n \rfloor$. Using Proposition 3.2, we obtain

$$\dim(G, m) \leq \dim(G, n) \leq \lfloor \log_2 m \rfloor$$

as claimed. ∎

Proposition 3.4. Let G have order n. For G for which $|\mathrm{Ord}(G, 2)| < m/2$, we have

$$\dim(G, m) \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

Proof. Let $k = \lfloor m/2 \rfloor$. If m is even, then $m = 2k$; if m is odd, then $m = 2k + 1$. Since less than $m/2$ elements in G have order 2, we may let

$$A = \{0\} \cup \bigcup_{i=1}^{k} \{a_i, -a_i\} \subset G$$

such that $a_i \neq -a_i \neq 0$ for all i and

$$\{a_i, -a_i\} \cap \{a_j, -a_j\} = \emptyset$$

for all $i \neq j$. If m is even, take A without $-a_k$. In either case, $|A| = m$.

Suppose that there exists a dissociated subset $D \subset A$ such that $|D| > k$. Since the smallest possible size of D is $k + 1$, if we choose $k + 1$ elements of A, we are forced to choose either 0 or both a_i and $-a_i$ for some $i \in \{1, \ldots, k\}$. In either case, $0 \in \Sigma D$, so we must have $|D| \leq k$, a contradiction. Therefore, $\dim A \leq k$, which directly implies that $\dim(G, m) \leq k$, as claimed. ∎

Note that these bounds are not generally sharp. For large m, Proposition 3.3 is better, but note that it is in terms of n, which loosens the bound.

Lower Bounds

In the next theorem, we introduce a lower bound for all G and m. First, we have a useful lemma, whose proof is omitted here.

Lemma 3.5. If G has invariant factorization

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r},$$

then the group

$$\mathbb{Z}_{n_{r-k+1}} \times \cdots \times \mathbb{Z}_{n_r}$$

is the largest subgroup of G with rank k.

This allows us to obtain the following result:

Theorem 3.6. Suppose that G has invariant factorization

$$G \cong \mathbb{Z}_{n_1} \times \cdots \times \mathbb{Z}_{n_r}.$$

Then for all m we have

$$\dim(G, m) \geq k$$

for the unique k such that

$$\prod_{i=r}^{r-k+2} n_i < m \leq \prod_{i=r}^{r-k+1} n_i.$$

Proof. Let $A \subseteq G$ have size m. Since

$$\prod_{i=r}^{r-k+2} n_i < m,$$

the size of A is strictly greater than that of the subgroup

$$\mathbb{Z}_{n_{r-k+2}} \times \cdots \times \mathbb{Z}_{n_r}.$$

By Lemma 3.5, this is the largest subgroup of G with rank $k-1$. Therefore, $\langle A \rangle$ has rank $r_A \geq k$. Now from Theorem 2.1 we have $\dim A \geq k$, and the claim follows. ■

From this we immediately obtain

Corollary 3.7. For all positive integers k and r, we have

$$\dim(\mathbb{Z}_k^r, m) \geq \lceil \log_k m \rceil.$$

Proof. Since $k^{i-1} < m \leq k^i$ for some $i \in \{1, \ldots, r\}$, we have $i \geq \lceil \log_k m \rceil$. The claim now follows from Theorem 3.6. ■

It should be noted that for $k > 3$, equality does not generally hold. For example, it is an immediate consequence from Theorem 2.1 that

$$\dim(\mathbb{Z}_k^r, k^r) \geq kr$$

for all r and $k \geq 4$, which is larger than the lower bound of r given here. We do not currently know much about other values of m.

Exact Values

It turns out that the bound in Corollary 3.7 is sharp for certain elementary abelian p-groups; in particular, $p = 2$ and $p = 3$.

Theorem 3.8. For all positive integers r and $m \leq 2^r$, we have

$$\dim(\mathbb{Z}_2^r, m) = \lceil \log_2 m \rceil.$$

Theorem 3.9. For all positive integers r and $m \leq 3^r$, we have

$$\dim(\mathbb{Z}_3^r, m) = \lceil \log_3 m \rceil.$$

The proofs of both theorems are similar, so here we prove both using a general construction.

Proof. Let $p \in \{2, 3\}$. Since p is prime, $\mathbb{Z}_p$ is a field and $\mathbb{Z}_p^r$ forms a vector space over $\mathbb{Z}_p$. Recall that for any $A \subseteq \mathbb{Z}_p^r$, the coefficients of sums in ΣA are given by $\lambda_i = \{0, 1, -1\}$. The scalars in $\mathbb{Z}_3^r$ are exactly λ_i. Also, since each element of $\mathbb{Z}_2^r$ is its own inverse, for $p = 2$ we only need to consider $\lambda_i = \{0, 1\}$, which are exactly the scalars in $\mathbb{Z}_2^r$. Therefore, for any $X \subseteq \mathbb{Z}_p^r$, we have an equivalence: X is dissociated if and only if the vectors in X are linearly independent.

We now use Proposition 3.2 to narrow down the cases; it suffices to prove that $\dim(\mathbb{Z}_p^r, p^k) = k$ and $\dim(\mathbb{Z}_p^r, p^k + 1) = k + 1$ for all positive integers $k \leq r$.

First, let $m = p^k$. From Theorem 3.7, we have $\dim(\mathbb{Z}_p^r, p^k) \geq k$, so it suffices to show the

existence of subset of $\mathbb{Z}_p^r$ of size p^k and dimension k. Let

$$A = \{0\}^{r-k} \times \mathbb{Z}_p^k.$$

It is clear to see that $\dim A = \dim \mathbb{Z}_p^k$. Now, it is a well known fact that if a vector space V is spanned by k vectors, and $l > k$, then any set of l vectors in V is linearly dependent. Since the standard basis for $\mathbb{Z}_p^k$,

$$\mathcal{B} = \{(e_1, \ldots, e_k) \mid e_i = 1 \text{ for some } i \in \{1, \ldots, k\},\ e_j = 0 \text{ for } j \neq i\},$$

spans $\mathbb{Z}_p^k$ and has size k, any subset of $\mathbb{Z}_p^k$ must have size at most k to be dissociated. It is clear that $0 \notin \Sigma\mathcal{B}$ since $\mathcal{B}$ is linearly independent, so $\dim \mathbb{Z}_p^k = k$. Therefore, we obtain $\dim A = k$ as desired.

Now let $m = p^k + 1$. Theorem 3.7 yields $\dim(\mathbb{Z}_p^r, p^k + 1) \geq k + 1$, so it again suffices to show existence of a subset of $\mathbb{Z}_p^r$ of size $p^k + 1$ with dimension $k + 1$. Write

$$A_0 = A \cup \mathcal{B}_0$$

where

$$\mathcal{B}_0 = \{1\} \times \{0\}^{r-1}.$$

Since $\mathcal{B} \cap \mathcal{B}_0 = \emptyset$, the set

$$(\{0\}^{r-k} \times \mathcal{B}) \cup \mathcal{B}_0 \subset A_0$$

has size $k + 1$ and is clearly dissociated.

Now, suppose that there exists a dissociated subset $D \subset A_0$ with $|D| > k + 1$. Since $\dim A = k$, D cannot contain more than k elements of A. But since we have $|A_0| = k + 1$, it is impossible to put two more elements of A_0 into D. Hence $|D| \leq k + 1$, a contradiction. From this, we obtain $\dim A_0 = k + 1$ as desired. ■

This proof does not work for $p \geq 5$; even though $\mathbb{Z}_p^r$ is a vector space for all values of p, the only groups where equivalence holds between dissociativity and linear independence are $\mathbb{Z}_2^r$ and $\mathbb{Z}_3^r$.

Using Proposition 2.2, we can easily get equality for a few related groups when m is maximal.

Proposition 3.10. For all positive integers r and k, we have

$$\dim(\mathbb{Z}_2^r \times \mathbb{Z}_{2k}, 2^{r+1}k) = \lfloor \log_2 2^{r+1}k \rfloor.$$

Proof. From Proposition 2.2, we have

$$r + \lfloor \log_2 2k \rfloor \leq \dim(\mathbb{Z}_2^r \times \mathbb{Z}_{2k}, k2^{r+1}) \leq \lfloor \log_2 k2^{r+1} \rfloor.$$

Since clearly

$$r + \lfloor \log_2 2k \rfloor = r + 1 + \lfloor \log_2 k \rfloor$$

and

$$\lfloor \log_2 k2^{r+1} \rfloor = \lfloor \log_2 2^r \cdot 2 \rfloor + \lfloor \log_2 k \rfloor = r + 1 + \lfloor \log_2 k \rfloor,$$

the bounds are equal and equality holds. ∎

Cyclic Groups

We now move on to cyclic groups $\mathbb{Z}_n$. In [2], B. Bajnok conjectured that for all positive integers n and $m \leq n$, the following equality holds:

$$\dim(\mathbb{Z}_n, m) = \lfloor \log_2 m \rfloor.$$

Using the code provided by Francis in [3], we have the following table of values that disagree with this conjecture:

G	m	$\dim(G, m)$	$\lfloor \log_2 m \rfloor$
$\mathbb{Z}_{17}$	$14, 15$	4	3
$\mathbb{Z}_{19}$	$14, 15$	4	3
$\mathbb{Z}_{22}$	15	4	3
$\mathbb{Z}_{23}$	$14, 15$	4	3
$\mathbb{Z}_{26}$	15	4	3
$\mathbb{Z}_{28}$	15	4	3
$\mathbb{Z}_{29}$	$14, 15$	4	3

Table 1: Values larger than the conjecture.

All other groups below $n = 30$ agree exactly for all values of m. In light of these results, I believe that the lower bound still holds.

I propose the following more modest conjecture:

Conjecture 3.11. For all positive integers n and $m \leq n$, we have

$$\dim(\mathbb{Z}_n, m) \geq \lfloor \log_2 m \rfloor.$$

I also believe that powers of two agree with Bajnok's conjecture.

Conjecture 3.12. For all positive integers n and $m \leq n$, we have

$$\dim(\mathbb{Z}_n, 2^k) = k.$$

It may also be more feasible to consider specific cyclic groups. For instance, the case when n is a power of two may be of particular interest.

4 Future Work

Future projects should explore the validity of Conjectures 3.11 and 3.12. Additionally, since the upper bounds of Proposition 3.4 and 3.3 are not always sharp, future work should look into the possibility of improving these bounds.

Furthermore, since Corollary 3.7 is only sharp for $k = 2$ and $k = 3$, future work should attempt to improve the lower bound for $k \geq 4$, and possibly other elementary abelian p-groups for $p \geq 5$.

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